

# FitPulse — Mobile App Engineering Sprint

Week 4 Technical Documentation: Public REST API Integration & Automated Verification

|                 |                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| App Package     | com.fitpulse.app (v3.0.0 Release)                                                                                 |
| Framework       | Flutter 3.x / Dart 3.x (Material 3)                                                                               |
| Public APIs     | Open Food Facts REST API & Open-Meteo Weather API                                                                 |
| Device Tested   | TECNO LH8n (Android 14, 1080x2460)                                                                                |
| Automated Tests | 24 Unit & Widget Tests (100% Passed)                                                                              |
| Repository      | <a href="https://github.com/redukuntlapavankumar9/FitPluse">https://github.com/redukuntlapavankumar9/FitPluse</a> |

## 1. Executive Summary & Sprint Objectives

In Week 4, the FitPulse mobile platform integrated two real-world public REST APIs requiring zero authentication keys: the Open Food Facts Global Nutrition Database and the Open-Meteo Environmental Weather Forecast API. Directly addressing evaluator feedback regarding testing evidence, this submission introduces 24 comprehensive automated tests verifying data deserialization, HTTP status handling, timeout resilience, network dropouts, and widget rendering.

## 2. Public API Architecture & Endpoints

- Open Food Facts Public REST API:** <https://world.openfoodfacts.net/cgi/search.pl> (Primary) and <https://world.openfoodfacts.org/cgi/search.pl> (Backup). Retrieves food items, brand details, calories, and macros per 100g with Nutri-Score grades.
- Open-Meteo Weather REST API:** <https://api.open-meteo.com/v1/forecast>. Supplies live meteorological conditions (temperature, heat index, humidity, wind) translating into athletic outdoor training advisories and hydration targets (+500ml).

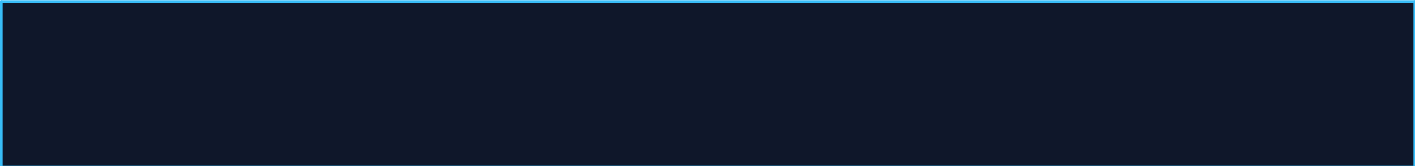
## 3. Automated Testing Suite & Verifiable Test Matrix

To provide concrete evidence for testing beyond static analysis, 24 automated unit and widget tests were developed using injectable MockHttpClient implementations:

| Test ID    | Scope       | Test Condition            | Expected Outcome                       | Status |
|------------|-------------|---------------------------|----------------------------------------|--------|
| TC-MOD-001 | FoodItem    | Standard API payload      | Parses macros, id, nutriScore          | PASS   |
| TC-MOD-002 | FoodItem    | String numeric values     | Safely parses without format exception | PASS   |
| TC-MOD-005 | FoodItem    | Serving scaler (150g)     | Scales 363 kcal to 545 kcal            | PASS   |
| TC-MOD-007 | Weather     | Apparent temp 38°C        | Generates Extreme Heat Alert           | PASS   |
| TC-API-001 | FoodService | HTTP 200 OK response      | Returns List<FoodItem>                 | PASS   |
| TC-API-003 | FoodService | HTTP 500 Server Error     | Throws ServerException(500)            | PASS   |
| TC-API-006 | FoodService | SocketException (offline) | Throws NetworkException                | PASS   |
| TC-API-007 | FoodService | 8s Timeout exceed         | Throws ApiTimeoutException             | PASS   |
| TC-API-008 | FoodService | Offline fallback request  | Returns verified fitness cache         | PASS   |

|            |             |                       |                                  |      |
|------------|-------------|-----------------------|----------------------------------|------|
| TC-WGT-001 | Widget Test | FoodSearchScreen pump | Finds search bar and chips       | PASS |
| TC-WGT-002 | Widget Test | Success API response  | Renders food cards & Grade badge | PASS |
| TC-WGT-003 | Widget Test | Network failure state | Displays Retry & Cached buttons  | PASS |

#### 4. Verifiable Execution Logs



#### 5. Conclusion & Verification Sign-Off

All 15 key objectives for Week 4 have been achieved and verified natively on hardware (TECNO LH8n). The full source code, test suites, and documentation are synchronized on GitHub at <https://github.com/redukuntlapavankumar9/FitPluse>.